MINIFLY RESEARCH / EXPERIMENT DESIGN / NOVEMBER 2026

Can two-step byte transitions rescue parity transfer?

Design only. Proposed E3 test on synthetic byte relations.

Question

The round-two memo reports 97.2% transfer for cyclic comparison on held-out distances and 2.1% for parity after comparison training. Test whether a correctly aligned two-step successor map makes parity learnable on untaught pairs. The inversion motivates the test and leaves open both representation and split effects.

Shared setup

Randomly assign 12 byte identities to a cycle independently in each world. Use the same background transitions, query format, fixed sparse expansion, and local delta readout in every arm. Learn successor maps with local TD updates from within-run transitions in the background stream. Two-step transitions traverse one of two alternating groups. Match feature dimensions, active units, readout size, map updates and teaching opportunities. Randomize the two answer-byte meanings independently in each world.

Arms

A. Aligned two-step. Concatenate a one-step and a two-step successor map.

B. Timescale control. Use two one-step maps with distinct discounts and the same storage.

C. Alignment control. Use one-step plus two-step maps. Randomly relabel the two-step observations only. Reject relabelings that preserve the same parity partition. Keep topology and update counts.

D. Feedback control. Use the aligned maps with parity writes disabled. Match exposure to answer bytes by presenting the same answer multiset in an unpaired sequence. Prevent answer bytes from updating the successor maps in all arms.

Learning and scoring boundary

The teaching marker is supplied with feedback. At a scored query, present only the prefix. Score the first answer and the learner-generated newline with all writes disabled. Use the same newline component across arms. No backpropagation, attention or content-addressed reads.

Evaluation and decision

Sealed holdouts

Primary. Withhold whole cyclic distances. Balance same and different parity in both training and test, and hold reverse orientations out together so an unordered pair cannot straddle the split. **Secondary.** Withhold one byte identity from every taught parity pair while keeping it in background transitions. Use 12 sealed worlds per suite, separate from engineering worlds.

Exposure and shortcut audit

Assert that no scored pair or its reverse received parity feedback. For the novel-item suite, audit both operand roles. Keep background transitions unlabelled and score before any query feedback. An evaluator-only reference must recover alternating groups from background transitions and infer the randomized answer mapping from permitted teaching pairs. Report exact-pair and unordered-pair lookup, constant, byte-frequency and operand-marginal baselines. Score answer-only accuracy and per-distance confusion alongside joint answer-plus-newline accuracy.

Decision rule

The aligned two-step arm must achieve at least 85% joint accuracy on the primary holdout and beat each matched representation control by at least 10 percentage points, with a paired 95% interval excluding zero for each contrast. Compute contrasts by world. The feedback control should remain near the balanced 50% answer-only baseline. Freeze settings, splits, metrics and the analysis rule before opening the final worlds. Report the secondary suite separately.

Interpretation

A pass supports a causal contribution from correctly aligned two-step structure on this synthetic relation. A tie with the timescale control leaves the specific two-step mechanism unresolved. If the reference passes and the native arms fail, probe whether the sparse features and readout expose the alternating groups. Record comparison-task retention separately if the prior comparison phase is included. Natural text, arithmetic and the MiniFly memory budget are outside this design.

Source notes

MiniFly Memo: Representation and Generalization, Round 2 (2026-11-03).
MiniFly Representation and Generalization Experiments (2026-11-03).